Adaptive Fourier Convolution Network for Road Segmentation in Remote Sensing Images

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***Abstract*— Segmentation of roads in remote sensing (RS) images is a challenging task due to the inhomogeneous intensity, nonconsistent contrast, and very cluttered background in remote sensing images. Recent approaches, mostly relying on convolutions or self-attention, make it difficult to extract weak and continuous road objects. Fourier neural operators (FNOs) provide another novel mechanism for capturing long-range and fine-grained features beyond self-attention. Based on it, we propose an adaptive Fourier convolution network (AFCNet) on the spatial–spectral domain for road segmentation in this article. The AFCNet is built on the pipeline of the classical U-Net model and its core is the proposed Fourier neural encoder (FNE), which is built on a feed-forward layer and a flexible Fourier convolutional structure composed of Fourier-domain pooling layers, asymmetric convolutions, squeeze-excitation inspired self-attention, and adaptive multiscale fusion (AMF) layers. Furthermore, we combine the FNE and bottleneck in ResNet to form a hybrid global–local feature representation scheme to capture the long and weak road objects in remote sensing images. The experiments on two public datasets, the Massachusetts Roads and DeepGlobe Road datasets, have shown that AFCNet worked with fewer parameters and outperformed most previous methods in terms of accuracy, precision, recall, and mean intersection over union (mIoU).**

***Index Terms*— Adaptive Fourier convolution (AFC), Fourier asymmetric convolution (FAC), Fourier neural encoder (FNE), road segmentation, U-Net.**

I. Introduction

ROAD segmentation in remote sensing (RS) images, which is essential for geographic information systems (GIS), urban road planning, and autonomous driving, is a fundamental research topic for cartography, traffic control, and intelligent transportation [1]. However, segmentation of roads from high-resolution remote sensing images still poses significant challenges [2] due to the inhomogeneous intensity, pixel mixing caused by low spatial resolution, geometric similarity of rivers and gullies, small-sized road objects interference, and very cluttered background in remote sensing images.

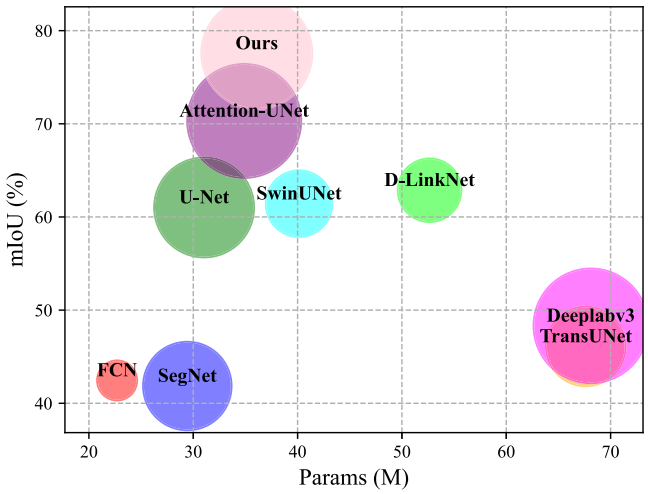


Fig. 1. Segmentation results by AFCNet and other baseline models on the DeepGlobe dataset: each bubble's area is proportional to the FLOPs of a variant in that model family. We demonstrate that a network model combining Fourier and convolutional operators can achieve better segmentation performance than the baseline models using only self-attention and regular convolutions. Moreover, our method achieves a parameter-efficient road segmentation model.

Deep neural networks have boosted road segmentation performance to a higher level than the traditional methods, which rely on expert knowledge and are usually based on handcrafted road features (e.g., shape, color, and texture). For instance, the classical convolution neural networks (CNNs), such as SegNet [3], U-Net [4], and their variants [5], [6], [7], [8], [9], have achieved promising results for road segmentation from remote sensing images. Some more recent versions, such as D-LinkNet [10] and ConDinet++ [11], adopt novel convolution layers (e.g., conditional dilated convolutions) to enhance fine-grained road features in remote sensing images with larger receptive field operators. Although these more up-to-date methods have achieved better performance, however, the inherent locality of these convolutional operations still limits the CNNs on capturing long and continuous road objects and weak edges in remote sensing images.

In contrast, the transformer model and its variants [5], [7], [12], [13] have shown great potential for simultaneous global and local feature extraction for different vision tasks. Notably, TransUNet [5], SwinUNet [7], and DCS-TransUperNet [13] have also been validated successfully for road extraction from remote sensing images. However, overly stacked self-attention

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blocks behave like low-pass filters [14] and suppress high-frequency signals and smoothen detailed feature maps, which means that these blocks are undesirable for tender object detection.

Beyond self-attention, the spectral convolution theorem [9], [15] provides another paradigm to design novel neural architectures to capture global features by locally operating in the Fourier domain. Several operators on the Fourier domain, such as fast Fourier convolution (FFC) [15], adaptive Fourier U-Net (AFU-Net) [9] and adaptive frequency filtering (AFF) [16], Fourier Image Transformer [17], and so on have shown more impressive performance and efficiency on diverse vision-related tasks. Similar Fourier-related neural operators, such as Fractional Fourier Image Transformer [18], Fractional Gabor convolutional neural networks [19], and so on are proposed for remote sensing topics. In remote sensing images, road areas have obvious low-frequency characteristics and show continuity. The areas beside the road have varying-frequency domain characteristics (e.g., much higher frequencies). Therefore, road segmentation based on remote sensing images is a very suitable application scenario for the Fourier domain. These Fourier-based works have shown fine-grained modulation learning with fractional convolution [18], Gabor filter [19], or other filters [9] on spectral-domain-based complex feature space has powerful capability to capture richer semantic road features in remote sensing images and can enhance the discrimination between anomalies and background.

Furthermore, recent work on asymmetric convolutions, for instance, asymmetric convolution network (ACNet) [20] and position aware circular convolution network (ParC-Net) [21], are powerful convolutional variants for capturing fine-grained features with efficient manners. Moreover, in the field of remote sensing image analysis, the dynamic frequency filtering [22] and modulation learning on spectral domain [9] usually separates the feature maps into low-frequency and high-frequency components or magnitude and phase components to grasp miniature feature differences, such as the rivers and gullies, vehicles on the road and the shadow of roadside buildings, by phase discrimination or attentive frequency filtering.

In this article, inspired by the Fourier neural operator (FNO), Transformer encoder, and asymmetric convolution, an adaptive Fourier convolution network (AFCNet), built on adaptive Fourier convolution (AFC) blocks and residual convolution blocks, following the U-Net pipeline, is proposed to provide a parameter-efficient and simultaneous local and global feature representation method to grasp more tender and continuous road features from remote sensing images (seen in Fig. 1). Specifically, the main contributions of this work can be summarized as follows.

1) We propose an AFCNet integrating the Fourier neural encoder (FNE) and residual convolution encoder blocks for road segmentation tasks in high-resolution remote sensing images.
2) We propose an FNE block combined with an FNO and feed-forward layer as a basic block for image global–local features representation.
3) We propose an AFC layer based on Fourier-domain pooling layers, asymmetric convolutions, squeeze-excitation-inspired meta-attention (MA), and adaptive multiscale fusion (AMF) mechanism to efficiently learn the fine-grained spectral features for capturing continuous and weak semantic features.
4) The proposed AFCNet achieves state-of-the-art (SOTA) performance on two publicly available datasets for road network extraction in remote sensing images.

II. Related Work

A. Road Segmentation From Remote-Sensing Images

In recent years, numerous innovative road network extraction methods have been proposed to address the challenge of automatic road segmentation from high-resolution remote sensing images. Traditional approaches include threshold-based segmentation [23] and edge detection-based methods [24]. However, traditional road segmentation methods based on thresholding and edge detection are often limited to specific scenarios, requiring manual tuning and expert knowledge. As deep learning has proven to be a more effective approach for segmentation tasks [25], an increasing number of remote sensing image segmentation tasks are now being addressed using deep-learning methods.

Compared to traditional methods, deep-learning-based approaches benefit from feature representation learning and parameter sharing, enabling automatic and effective road identification. In recent years, the rapid development of CNNs has provided sound technical support for semantic segmentation. For instance, Gao et al. [26] introduced a CNN-based multifeature pyramid network framework by expanding the receptive field to capture more detailed information about narrow roads. Zhou et al. [10] used a pretrained encoder from the LinkNet [27] architecture as their backbone and combined it with dilated convolution layers to further address the connectivity and long-range properties of roads. These feature pyramid-based methods have demonstrated superior performance in road segmentation from remote sensing images. Additionally, CNNs have shown excellent performance in remote sensing image segmentation, particularly those with the encoder–decoder structure like U-Net [4]. For example, Liu et al. [28] combined residual learning and dilated convolution in the U-Net structure to expand the receptive field of operators without reducing the resolution of feature maps, thereby simplifying network training while enhancing segmentation performance. Yang et al. [11] improved the commonly encountered encoder–decoder structure in semantic segmentation models by utilizing conditional dilated convolution blocks to enlarge the network's receptive field and enhance semantic feature extraction capabilities. Zhang et al. [13] proposed a features fusion module (FFM) by combining the architecture of UperNet [29] with the multiscale vision Transformer [30] to fuse coarse-grained and fine-grained features from two encoder subnetworks, enabling more accurate road segmentation.

However, the limited receptive field in the convolution layers poses challenges in capturing long-range spatial

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dependencies and achieving global semantic interpretation in images. To accurately extract the road objects in remote sensing images, enhancing the representation capability of operators is essential for detecting these tortuous, thin, and long objects from complex backgrounds. Although CNNs can enlarge the receptive field by stacking convolution layers, downsampling, or dilated convolutions, these approaches might result in complexity increasing and a higher risk of overfitting.

B. Asymmetric Convolutions

Asymmetric or separable convolution is a technique used for model compression and acceleration by decomposing existing square convolutions into asymmetric ones. Since a rank-1 matrix can be decomposed into the outer product of a column vector and a row vector, a rank-1 2-D convolution kernel belongs to asymmetric convolution kernels, which can be equivalently transformed into a series of 1-D convolutions through the decomposed parallel and serial operations. Actually, asymmetric convolutions are often applied in various lightweight deep neural networks. For instance, in the Inception-v3 [31], a 7×7 convolution kernel is decomposed into a 1×7 convolution kernel and a 7×1 convolution kernel. The concatenation of these two kernels is equivalent to a two-layer network with the same receptive field as the 7×7 convolution kernel but with significantly reduced computational complexity. In the depthwise separable convolutions [32], [33], a standard convolution operation is decomposed into a sequential combination of a depth-wise convolution (DWConv) and a point-wise convolution, not only increasing representational efficiency, but also consuming fewer parameters and less computational burden. EDANet [34] employs a similar approach to decompose the 3×3 convolution, reducing the parameter count by 33% and achieving real-time and efficient segmentation. ACNet [20] enhances the effect of the 3×3 convolution in the horizontal and vertical directions by using asymmetric convolution structures composed of 3×1, 1×3, and 3×3 convolutions, highlighting local key features and accelerating training efficiency.

Up to date, asymmetric convolutions are utilized on the spatial domain of images and can efficiently extract richer features for several image-related tasks and are seldom applied on the frequency domain.

C. Fourier Convolutions and Fourier Embeddings

The Fourier transform is an important tool for time–frequency analysis [35], providing an effective method for image feature representation in the frequency domain. Recent developments in Fourier convolution and Fourier-embedded learning methods have demonstrated the richer feature representation capabilities with complex numbers and global feature representation capabilities by Fourier convolutions. Moreover, several learnable frequency-domain filters have shown great achievements in different image-related tasks. For instance, Wang et al. [36] designed a deep residual network with frequency-domain convolution, which could suppress image noises and artifacts when reconstructing images from undersampled data. Lee-Thorp et al. [37] replaced the self-attention in vision Transformers with a simple, unparameterized Fourier transform that "mixes" input tokens, addressing the computational-complexity issue caused by token size in high-resolution settings. The global filter network [38] using a global filter on the Fourier domain showed strong competitiveness in terms of efficiency, generalization, and robustness compared to Transformer and conventional convolution models. Guibas et al. [39] introduced adaptive FNOs to represent the image features with sparse spectrum on the Fourier domain. The use of sparse regularization in these operators significantly reduced the computational requirements while achieving superior performance compared to self-attention mechanisms. A Fourier domain encoding method in Fourier image Transformer [17] is proposed to predict a higher resolution output for image completion with an autoregressive mechanism. What is more, Fourier-embedded convolutional neural networks have been successfully applied to remote sensing. For instance, in [18], the fractional Fourier image transformer is proposed as a backbone network to extract both global and local contexts for multimodal remote sensing data classification from fractional and Fourier domains. Similarly, in [19], the fractional Gabor convolutional layer is proposed to extract multiscale, multidirectional, and semantic change features for multisource remote sensing data classification.

Since complex numbers offer more detailed feature representation by phases compared to real numbers, in recent years, there has been increasing success in employing complex-valued networks with Fourier embedding for image-related tasks. Simply separating the real and imaginary parts or considering only magnitude and phase angles would lose the inherent relationships within complex numbers. Therefore, complex network [40] represents the real and imaginary parts of complex numbers as two logically distinct real-value vectors, using real-value arithmetic operations to simulate complex operations. This approach ensures that input neurons with similar phase angles are added synchronously in complex arithmetic while adding asynchronous neurons leads to destructive interference and thus subtraction is used to counteract the interference. Gu and Ding [41] transformed the VGG model from a real-valued model in the time domain to a complex-valued model in the frequency domain, discovering improved stability and faster convergence in image recognition tasks. Li et al. [42] proposed frequency-domain split-transform merge and feature recalibration methods based on complex-valued convolution to further focus on the extracted features, resulting in improved image classification accuracy. Yang and Liu [9] introduced modulation learning in the Fourier domain, achieving fine-grained modulation on the Fourier spectrum through global, phase, and magnitude modulation modules, leading to significant progress in road segmentation in remote sensing images.

Compared to time-domain deep-learning models, frequency-domain models based on Fourier transform can more efficiently consider global information and effectively integrate

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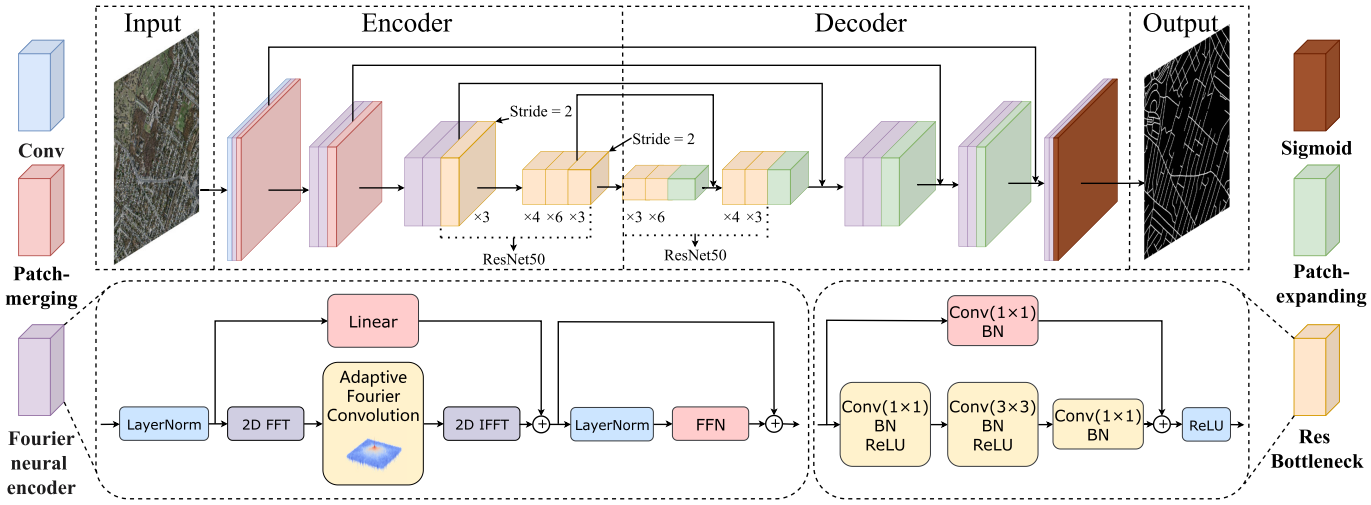


Fig. 2. Architecture of AFCNet.

local and nonlocal receptive fields, thereby further boosting the model's performance.

III. Proposed Method

A. Overall Architecture

To address the locality limitations of traditional CNNs for road extraction in remote sensing images, an AFCNet for road extraction from high-resolution remote sensing images is proposed here. The architecture of the AFCNet is depicted in Fig. 2.

Overall, the AFCNet follows the pipeline of U-Net and consists of the encoder and decoder parts. The RGB image is taken as the input of the AFCNet. Its encoder starts with a stem block that has a convolution layer with the kernel size of 3 × 3 and a stride of 1. Each layer is followed by a batch normalization (BN) and a Gaussian error linear unit (GELU) activation function. The encoder aims to project and resample the image RGB space to high-dimensional feature space from remote sensing images to extract the long and tender road features. Initially, the input image is projected into high-dimensional feature space with a stem block composed of conventional convolution layers. Subsequently, features are further reprojected into higher-dimensional space through four levels of encoder modules. In this part, a multiscale hybrid encoder scheme is utilized to refine local–global features. Specifically, on high-resolution features, we use the proposed FNE block in the spatial–spectral domain with good global feature modeling capability for long and continuous feature extraction. On low-resolution features, we use faster and conventional convolutions under spatial domain with small kernels in ResNet [43] for fine-grained and weak feature extraction. Correspondingly, the decoder aims to reconstruct the predicted results from the high-dimensional features, and its symmetrical four-level decoder block is configured with the same spatial–spectral domain feature extraction block and conventional convolution as in the encoder. The patch-merging module in the encoder and patch-expanding module in the decoder are applied for downsampling and upsampling, respectively, to reduce and expand the spatial dimension of the feature space. In our model, the output feature maps of each stage of the encoder are directly connected with the decoder across corresponding stages. Channel concatenation is performed to fuse shallow features and deep ones across the corresponding encoder and decoder stages. Finally, a convolution layer, whose kernel size is 1 × 1, with a Sigmoid classifier is used to infer the probability of road objects and background.

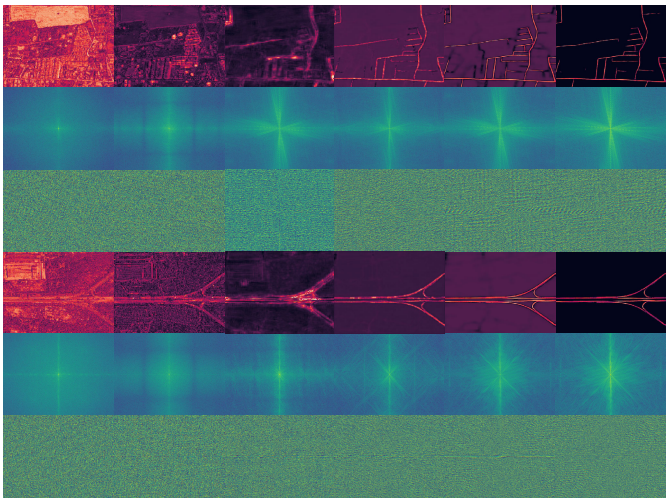


Fig. 3. Feature maps are visualized spatial–spectral features of the FNE block in the encoder and decoder parts.

The visualized spatial–spectral feature representation of the FNE block of AFCNet is shown in Fig. 3, and the exhibited feature maps from left to right are features in the first three FNE layers in the encoder and the last three FNE layers in the decoder. From top to bottom, the maps are features on the spatial domain, amplitude maps, and phase maps of the spectral domain, respectively. The synchronous generating process of spatial and spectral features in our AFCNet can be analyzed from the visualized results. It can be noticed from the amplitude maps that the energy more concentrates on some

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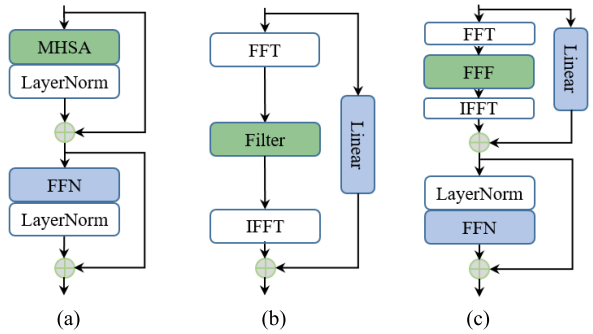


Fig. 4. Structure of FNE. (a) Transformer encoder. (b) FNO. (c) Our FNE.

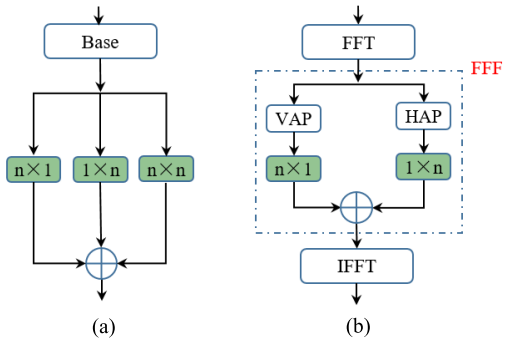


Fig. 5. Structure of FAC. "VAP" refers to vertical average pooling and "HAP" refers to horizontal average pooling. (a) Asymmetric conv. (b) Fourier asymmetric conv.

specific frequencies with the reduced scale in the encoder part, which is consistent with the representations of spatial feature maps where the high-level encoder layer reflects more on high-frequency information.

B. Fourier Neural Encoder

The blooming of the Transformer has shown that the Transformer encoder [40] consisting of self-attention and feed-forward layer [seen in Fig. 4(a)] is powerful for most deep-learning tasks benefiting from its global–local feature representation capability under the spatial domain. Notably, besides self-attention, the spectral convolution theorem [15], [44] provides another path to capture global information and integrate nonlocal receptive fields by point-wise updates in the spectral domain in a more efficient manner, thereby further enhancing the model's performance, especially in capturing more fine-grained features from complex backgrounds. For instance, in the FNO [45] [seen in Fig. 4(b)], some specific filters on the Fourier domain, for example, the global filter on the whole spectral feature space [38], adaptive FNO [39] on the high-energy concentrated sparse spectral representation have shown great potential for vision-related tasks.

Meriting from the elegant structure of the Transformer encoder and the charming performance of FNOs for several vision tasks, we rethink the encoder's structure and rebuild a novel FNE block on the spatial–spectral domain, which is depicted in Fig. 4(c). The FNE consists of an FNO based on some flexible Fourier filters (FFFs) on the spectral domain, a feed-forward network (FFN), and a linear transform layer (Linear) on the spatial domain. The FFT and IFFT in Fig. 4(b) and (c) are 2-D fast Fourier transform and 2-D inverse fast Fourier transform [15], respectively. In practice, 2-D discrete Fourier transform (DFT) and 2-D inverse DFT (IDFT) operators are often used for image-related tasks. In this work, the proposed FNE block which would replace the classical Transformer encoder block can be reformulated as

$$\hat{\mathbf{X}}'_l = \mathcal{F}^{-1}\big(\mathrm{FFF}\big(\mathcal{F}\big(\hat{\mathbf{X}}_{l-1}\big)\big)\big) + \mathrm{Linear}\big(\hat{\mathbf{X}}_{l-1}\big) \quad (1)$$

$$\hat{\mathbf{X}}_l = \mathrm{FFN}\big(\mathrm{LN}\big(\hat{\mathbf{X}}'_l\big)\big) + \hat{\mathbf{X}}'_l \quad (2)$$

where $\hat{\mathbf{X}}_l \in \mathbb{R}^{C\times H\times W}$ is the lth-layer feature and $\mathcal{F}(\cdot)$ and $\mathcal{F}^{-1}(\cdot)$ are DFT and IDFT, respectively.

C. Fourier Asymmetric Convolution Layer

Several FFFs have been applied for vision-related tasks, for instance, the global filter on the whole spectral feature space [38], adaptive FNO [39] on the high-energy concentrated sparse spectral representation have shown great potential for vision-related tasks, but they require a large number of trainable parameters, which would significantly increase the computational burden and memory consumption. Thus, we need more efficient and lightweight versions that could be deployed on resource-constrained devices. Several convolution variants [20], [21], [33] have been proven to be more efficient ways to replace standard convolutions. Among them, asymmetric convolution is one of the flexible variants to reduce computational demand. An asymmetric convolution usually splits a square kernel into asymmetric smaller ones. For instance, in the ACNet [20] [seen in Fig. 5(a)], an $n \times n$ standard convolution kernel is decomposed into three parallel layers with an $n \times n$ skeleton kernel, $1 \times n$, and $n \times 1$ asymmetric kernels to improve the performance of various models but requiring no extra computations anymore.

Asymmetric convolutions or separables, for example, the ACNet [20] and Xception algorithm [32], have been proven great when optimizing the model for smaller size or higher speed compromising least with accuracy. However, spatially separable convolutions tend to underfit, and not every kernel can be divided into two kernels in the case of spatial formation due to the possible existence of a nonpositive definite kernel matrix. Based on it, a flexible separating scheme on the Fourier domain, including projecting the spectra from orthogonal directions and running asymmetric convolutions on corresponding directions, has been proposed to balance the tradeoff between reducing the number of trainable parameters and maintaining effective feature extraction.

As shown in Fig. 5(b), we bridge the asymmetric convolution with Fourier convolution to rebuild a flexible Fourier asymmetric convolution (FAC) layer, which consists of three compact parts: vertical average pooling (VAP) and horizontal average pooling (HAP) for spectral projecting, asymmetric convolutions for extracting spectral-related fine-grained features on frequency domain and a feature aggregation operation.

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1) Vertical and Horizontal Average Pooling: The average pooling (AP) is a highly effective way to aggregate feature information. One possible explanation for the effectiveness of AP is considering each feature vector as representing a different semantic entity and AP as a convex combination of them [46]. Besides classical global AP (GAP) and global max-pooling (GMP) [47], both VAP and HAP are novel aggregating methods, respectively, on the Fourier domain in this work. Specifically, consider the composite function family $e = g \circ \mathbf{X}$ such that g is a pooling function and $\mathbf{X} \in \mathbb{C}^{C\times H\times W}$ is a complex feature tensor. We assume a further structure over the function g and feature $\mathbf{X}$. The VAP function maps the input space $\mathbf{X} \in \mathbb{C}^{C\times H\times W}$ into $\mathbf{X}_V \in \mathbb{C}^{C\times H\times 1}$, where H and W are spatial dimensions and C is the channel dimension. Moreover, VAP$(\cdot)$ performs vertical averaging pooling as

$$\mathbf{X}_V = \mathrm{VAP}(\mathbf{X}) = \frac{1}{|C_h|}\sum_{w\in[0,W-1]} \mathbf{X}_{[:,:,w]} \tag{3}$$

where $\mathbf{X}_{[:,:,w]} \in \mathbf{X}$ and $|C_h|$ indicates the hth row of the total number of elements across the spectral feature map.

Similarly, the HAP function, short for HAP$(\cdot)$, maps the input space $\mathbf{X} \in \mathbb{C}^{C\times H\times W}$ into $\mathbf{X}_H \in \mathbb{C}^{C\times 1\times W}$, and it can be formulated as

$$\mathbf{X}_H = \mathrm{HAP}(\mathbf{X}) = \frac{1}{|C_w|}\sum_{h\in[0,H-1]} \mathbf{X}_{[:,h,:]} \tag{4}$$

where $\mathbf{X}_{[:,h,:]} \in \mathbf{X}$ and $|C_w|$ indicates the wth column of the total number of elements across the spectral feature map.

2) Fourier Asymmetric Convolutions: The feature tensor is first transformed into two compressed tensors in the horizontal and vertical directions using APs (e.g., VAP and HAP), which replaces the full tensor to reduce the frequency dimensionality of spectral features. Subsequently, the compressed feature map is convolved with individual asymmetric kernels along the horizontal and vertical axis to extract representative features. The diagram of asymmetric convolution on the Fourier domain is shown in Fig. 5(b), which can be formulated as

$$\mathbf{X}_F^H = \mathrm{ConvH}(\mathbf{X}_H) \tag{5}$$

$$\mathbf{X}_F^V = \mathrm{ConvV}(\mathbf{X}_V) \tag{6}$$

where $\mathbf{X}_F^H \in \mathbb{C}^{C\times 1\times W}$ and $\mathbf{X}_F^V \in \mathbb{C}^{C\times H\times 1}$ denote the complex features along the horizontal and vertical axis, respectively, and ConvH$(\cdot)$ and ConvV$(\cdot)$ denote the asymmetric complex-valued convolution modules with kernel sizes of $1\times n$ and $n\times 1$, respectively. Each convolution module consists of two asymmetric complex-valued DWConvs [40], a regularization layer (e.g., BatchNorm [40]), and an activation function layer (e.g., Hardwish [48]), in the order of DWConv$\rightarrow$BatchNorm$\rightarrow$Hardwish$\rightarrow$DWConv. Those sequential DWConvs are stacked together for feature squeezing and excitation and their channels are following the configuration of $C\rightarrow C/2\rightarrow C$.

3) Feature Aggregation: Classical feature aggregation methods include tensor addition or tensor concatenation. Here, the usually used feature fusion scheme can be formulated as

$$\mathbf{X}_F = \bar{\mathbf{X}}_F^V + \bar{\mathbf{X}}_F^H \tag{7}$$

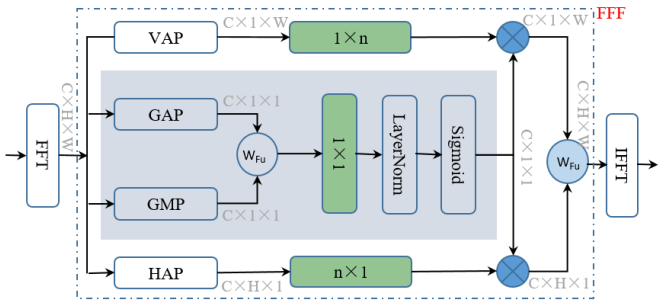


Fig. 6. Structure of AFC.

where $\mathbf{X}_F \in \mathbb{C}^{C\times H\times W}$ is the output tensor, and $\bar{\mathbf{X}}_F^H$ and $\bar{\mathbf{X}}_F^V$ are complex features after the tensor broadcasting operation with PyTorch.

D. Adaptive Fourier Convolution Layer

Based on the above FAC layer, we further upgrade it with a channel-wise MA layer and an AMF layer to an adaptive version (shown in Fig. 6). Specifically, our AFC layer can adaptively enhance the semantic information with attentive features by MA layers, and it can adaptively recalibrate those multiscale feature maps by learning the fusion parameters, which would help our model to better align features at different scales and fuse more consistent semantic information. Moreover, the hybrid regularization strategy, where the BatchNorm and Hardwish are used in the FAC layer and the LayerNorm and Sigmoid are used in the MA layer, is adopted for model robustness and generalization. The squeeze-excitation attention mechanism [49] tells us content-aware global contextual modeling and channel adaptive re-weighting can efficiently enhance the semantic information and suppress the nonsemantic features. In our work, an adaptive global contextual modeling mechanism, which combines GAP and GMP adaptively, is used. Specifically, the GAP focuses on the basic spectral feature and suppresses the strongest and weakest spectra. Meanwhile, the GMP is designed to capture the most salient spectral while ignoring the spectral background. Subsequently, we employ the channel-wise MA layer to re-weight the channel information to corresponding separated spectral branches. Specifically, a layer normalization [50] LN$(\cdot)$ for normalization on the channel dimension following a 1×1 convolution is used for facilitating the model's generalization, and a Sigmoid activation function $\sigma(\cdot)$ to form nonlinear features to suppress the nonsemantic features, meanwhile enhance the semantic channels.

1) Global Average Pooling: GAP is a special case of AP, while complex GAP smooths all elements across the spectral feature map. It can be formulated as

$$\mathbf{X}_{\mathrm{GAP}} = \mathrm{GAP}(\mathbf{X}) = \frac{1}{|C|}\sum_{(h,w)\in[H,W]} \mathbf{X}_{[:,h,w]} \tag{8}$$

where $\mathbf{X}_{\mathrm{GAP}} \in \mathbb{C}^{C\times 1\times 1}$ denotes the AP results on the entire spectral feature map, $\mathbf{X}_{[:,h,w]} \in \mathbb{C}^{C\times 1\times 1}$ represents the complex-valued element located at (h, w) in the feature map,

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and $|C|$ indicates the total number of elements on the feature map.

2) Global Max-Pooling: Similarly, it operates on the entire spectral map so that the maximum real and imaginary components of complex features on the channel dimension will be selected. It can be expressed as

$$\mathbf{X}_{\mathrm{GMP}} = \mathrm{GMP}(\mathbf{X}) = \max_{(h,w)\in[H,W]} \mathbf{X}_{[:,h,w]} \tag{9}$$

where $\mathbf{X}_{\mathrm{GMP}} \in \mathbb{C}^{C\times 1\times 1}$ denotes the GMP results on the space of complex-valued feature map, $\mathbf{X}_{[:,h,w]} \in \mathbb{C}^{C\times 1\times 1}$ represents the element located at (h, w) in the frequency feature map, and $\max(\cdot)$ is an operator to select the maximum real and imaginary parts of complex features individually.

3) Meta-Attention Layer: The above two types of global features are fused to recalibrate the channel features and enable the separated convolution layers to enhance the semantic channel components and meanwhile suppress the nonsemantic ones. The squeeze-excitation-like channel-wise MA layer is defined as

$$\mathbf{X}_F^C = \sigma\left(\mathrm{LN}\left(\mathrm{Conv}\left(\mathbf{W}_{\mathrm{Fu}}^1 \odot \mathbf{X}_{\mathrm{GMP}} + \mathbf{W}_{\mathrm{Fu}}^2 \odot \mathbf{X}_{\mathrm{GAP}}\right)\right)\right) \tag{10}$$

where $\mathbf{W}_{\mathrm{Fu}}^1 \in \mathbb{C}^{C\times 1\times 1}$ and $\mathbf{W}_{\mathrm{Fu}}^2 \in \mathbb{C}^{C\times 1\times 1}$ are learnable parameters to adaptively align of channel features, $\mathrm{Conv}(\cdot)$ is a complex convolution with 1×1 kernel, $\mathrm{LN}(\cdot)$ is the complex LayerNorm [50], and $\sigma(\cdot)$ is a complex Sigmoid operator [40].

4) Adaptive Multiscale Fusion: Classical feature aggregation methods include tensor addition or tensor concatenation, and there are inherent limitations in these methods: 1) the features at different scales could not be fully aligned and 2) the features with different dimensions could not be semantically aggregated with a more consistent manner. Aligning two branches of separated features in vertical and horizontal directions aims to learn more consistent features and further aggregate more rich semantic information of tender road objects. Therefore, we propose an AMF module based on parameter learning for feature aggregation.

By learning the fusion parameters, our fusion layers can adaptively recalibrate those multiscale feature maps, which would help our model to better align features on different scales and to fuse more consistent semantic information. The multiscale features after asymmetric convolutions can be adaptively aligned and further refined by the MA layer. Then, our model can adaptively calibrate the fusion weights of the multiscale features and refine the separated features by channel-attentive features, which not only enhance the adaption capability of layers to feature noise and outliers, but also refine the semantic features of weak road objects, and can finally enhance its feature precision and representation capabilities. The proposed AMF layer (seen in Fig. 6) can be formulated as

$$\mathbf{X}'_F = \bar{\mathbf{W}}_{\mathrm{Fu}}^1 \odot \left(\bar{\mathbf{X}}_F^C \odot \bar{\mathbf{X}}_F^V\right) + \bar{\mathbf{W}}_{\mathrm{Fu}}^2 \odot \left(\bar{\mathbf{X}}_F^C \odot \bar{\mathbf{X}}_F^H\right) \tag{11}$$

where $\odot$ denotes complex element-wise multiplication, $\mathbf{X}'_F \in \mathbb{C}^{C\times H\times W}$ is the output tensor, $\mathbf{W}_{\mathrm{Fu}}^1 \in \mathbb{C}^{C\times 1\times W}$ and $\mathbf{W}_{\mathrm{Fu}}^2 \in \mathbb{C}^{C\times H\times 1}$ are learnable parameters to adaptively calibrate the weights of vertical and horizontal features, and $\bar{\mathbf{X}}_F^C$, $\bar{\mathbf{X}}_F^H$, $\bar{\mathbf{X}}_F^V$, $\bar{\mathbf{W}}_{\mathrm{Fu}}^1$, and $\bar{\mathbf{W}}_{\mathrm{Fu}}^2$ are complex features after the tensor broadcasting operation with PyTorch.

E. Residual Convolution Layer

Despite the global feature representation capabilities to capture the tortures and long road objects based on Fourier convolution layers, fine-grained features on a small scale are essential for tender and weak road object detection. Therefore, classical small-kernel convolution blocks, for example, the bottleneck in the ResNet [43], are widely used for semantic segmentation and are utilized as fundamental encoder and decoder blocks in our AFCNet as well. It also releases the risk of vanishing gradient by introducing an identity shortcut connection.

In the classical ResNet-50 [43], it is divided into four stages, and they consist of three, four, six, and three bottlenecks, respectively. In the encoder of small-scale feature extraction, the bottleneck of ResNet-50 is adopted for fine-grained feature extraction. Moreover, the residual bottleneck block in our model is a variant of residual blocks (seen in Fig. 2), where we utilize grouped convolutions to reduce the number of parameters.

F. Loss Function

Considering the significantly small proportion of road objects in the whole remote sensing images, a weighted loss function is adopted for full supervision during the training phase, which is composed of a softDice loss function and a softCPclDice loss function ever used in [51] and focused on preserving the road connectivity. The loss function L_α is defined as follows:

$$L_\alpha = (1-\alpha)(1-\mathrm{softDice}) + \alpha(1-\mathrm{softCPclDice}) \tag{12}$$

$$\mathrm{softDice} = 2 \times \frac{M_D \cdot M_G}{M_D + M_G} \tag{13}$$

$$\mathrm{softCPclDice} = 2 \times \frac{C_{\mathrm{prec}}(S_D, M_G) \times C_{\mathrm{secs}}(S_G, M_D)}{C_{\mathrm{prec}}(S_D, M_G) + C_{\mathrm{secs}}(S_G, M_D)} \tag{14}$$

$$C_{\mathrm{prec}}(S_D, M_G) = \frac{|\ S_D \circ M_G\ |}{|\ S_D\ |} \tag{15}$$

$$C_{\mathrm{sens}}(S_G, M_D) = \frac{|\ S_G \circ M_D\ |}{|\ S_G\ |} \tag{16}$$

where α is the weight coefficient of two components, M_G is the ground-truth mask, M_D is the detected segmentation mask, which are two binary masks, and S_D represents the detected skeleton of a predicted mask M_D, while S_G represents the skeleton of a ground-truth mask M_G, and $\circ$ denotes the Hadamard product and $\cdot$ denotes the pixel-wise multiplication.

IV. Datasets and Performance Metrics

A. Datasets

Our proposed AFCNet model is trained and evaluated on two public datasets: the Massachusetts Road dataset [52] and the DeepGlobe Road dataset [53]. The Massachusetts Road dataset consisted of aerial images of the state of Massachusetts. It contains 1108 images for training, 49 images for testing, and 14 images for validation. The resolution of

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each image is 1500 × 1500. The DeepGlobe dataset comes from the 2018 CVPR Road Extraction Challenge, consisting of images captured over Thailand, Indonesia, and India, covering a variety of urban, suburban, and rural regions. This dataset is randomly divided into 4980 images for training, 624 for testing, and 626 for validation. It contains 4980 images for training and 1246 images for testing. The resolution of each image is 1024 × 1024.

B. Performance Metrics

In this article, to evaluate the performance of the proposed network in road extraction tasks, the widely accepted metrics as in [54], precision, recall, F-score, and mean intersection over union (mIoU) are used for performance evaluation. The pixels in the segmentation image are divided into true positive (TP), false positive (FP), true negative (TN), and false negative (FN). Then four metrics are calculated as follows:

$$\text{mIoU} = \frac{1}{2} \times \left(\frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} + \frac{\text{TN}}{\text{TN}+\text{FP}+\text{FN}} \right) \quad (17)$$

$$\text{Accuracy} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \quad (18)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \quad (19)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}}. \quad (20)$$

Moreover, the efficiency analysis in terms of floating-point operations per second (FLOPS) and frames per second (FPS) is also used for computational complexity and inference speed evaluation, and all models are evaluated on a single Nvidia 3090 graphics processing unit (GPU).

V. Experimental Analysis

A. Implementation Details

In Table I, we introduced the detailed definition of our proposed AFCNet. Specifically, we listed the detailed definition of each block about its input size ("In. S." in short), output size ("Out. S." in short), input channel ("In. Ch." in short), output channel ("Out. Ch." in short), and corresponding parameters at each scale in the encoder and decoder stages, supposing that the size of the input image is $H \times W$.

During training, a data augmentation strategy has been utilized to expand the training set, and image transformations, including rotation, scaling, flipping, and mirroring, are also applied to both datasets for data augmentation purposes. For instance, some images are rotated randomly with 90°, 180°, and 270°, and some images are flipped along the vertical and horizontal axis, and scaling the original images to 0.5, 0.75, and 1.5 times has also been used for data augmentation. Moreover, each image in the Massachusetts Road dataset is cropped into subimages of fixed size 448 × 448 under the stride of 448 pixels, and the remaining parts of cropped images are filled with zero padding. During inference, output results will be mosaiced to the original size.

Our proposed model is implemented by PyTorch 1.8 on a single NVIDIA 3090 GPU. The training batch size is set to 4 and the stochastic gradient descent (SGD) optimizer is utilized as the learning rate scheme. The initial learning rate is $\lambda = 10^{-2}$ and linearly decayed to $\lambda = 10^{-6}$ after 160 000 iterations.

TABLE I
AFCNet Definition Details

Stage	Scale	Block Name	In. S.	Out. S.	In. Ch.	Out. Ch.
Encoder	Scale 1	Conv 3 × 3	$H \times W$	$H \times W$	3	64
		FNE	$H \times W$	$H \times W$	64	64
		Patch-Merging	$H \times W$	$\frac{H}{2} \times \frac{W}{2}$	-	-
	Scale 2	FNE	$\frac{H}{2} \times \frac{W}{2}$	$\frac{H}{2} \times \frac{W}{2}$	64	128
		FNE	$\frac{H}{2} \times \frac{W}{2}$	$\frac{H}{2} \times \frac{W}{2}$	128	128
		Patch-Merging	$\frac{H}{2} \times \frac{W}{2}$	$\frac{H}{4} \times \frac{W}{4}$	-	-
	Scale 3	FNE	$\frac{H}{4} \times \frac{W}{4}$	$\frac{H}{4} \times \frac{W}{4}$	128	256
		FNE	$\frac{H}{4} \times \frac{W}{4}$	$\frac{H}{4} \times \frac{W}{4}$	256	256
		Bottleneck × 3	$\frac{H}{4} \times \frac{W}{4}$	$\frac{H}{8} \times \frac{W}{8}$	256	256
	Scale 4	Bottleneck × 4	$\frac{H}{8} \times \frac{W}{8}$	$\frac{H}{8} \times \frac{W}{8}$	256	512
		Bottleneck × 6	$\frac{H}{8} \times \frac{W}{8}$	$\frac{H}{8} \times \frac{W}{8}$	512	512
		Bottleneck × 3	$\frac{H}{8} \times \frac{W}{8}$	$\frac{H}{16} \times \frac{W}{16}$	512	1024
Decoder	Scale 5	Bottleneck × 3	$\frac{H}{16} \times \frac{W}{16}$	$\frac{H}{16} \times \frac{W}{16}$	1024	1024
		Bottleneck × 6	$\frac{H}{16} \times \frac{W}{16}$	$\frac{H}{16} \times \frac{W}{16}$	1024	1024
		Patch-Expanding	$\frac{H}{16} \times \frac{W}{16}$	$\frac{H}{8} \times \frac{W}{8}$	-	-
	Scale 4	Bottleneck × 4	$\frac{H}{8} \times \frac{W}{8}$	$\frac{H}{8} \times \frac{W}{8}$	1024	512
		Bottleneck × 3	$\frac{H}{8} \times \frac{W}{8}$	$\frac{H}{8} \times \frac{W}{8}$	512	512
		Patch-Expanding	$\frac{H}{8} \times \frac{W}{8}$	$\frac{H}{4} \times \frac{W}{4}$	-	-
	Scale 3	FNE	$\frac{H}{4} \times \frac{W}{4}$	$\frac{H}{4} \times \frac{W}{4}$	512	256
		FNE	$\frac{H}{4} \times \frac{W}{4}$	$\frac{H}{4} \times \frac{W}{4}$	256	256
		Patch-Expanding	$\frac{H}{4} \times \frac{W}{4}$	$\frac{H}{2} \times \frac{W}{2}$	-	-
	Scale 2	FNE	$\frac{H}{2} \times \frac{W}{2}$	$\frac{H}{2} \times \frac{W}{2}$	256	128
		FNE	$\frac{H}{2} \times \frac{W}{2}$	$\frac{H}{2} \times \frac{W}{2}$	128	128
		Patch-Expanding	$\frac{H}{2} \times \frac{W}{2}$	$H \times W$	-	-
	Scale 1	FNE	$H \times W$	$H \times W$	128	64
		FNE	$H \times W$	$H \times W$	64	64
		Conv 1 × 1	$H \times W$	$H \times W$	64	2
Final			$H \times W$	$H \times W$	2	2

B. Comparison With the State-of-the-Art

The comparison experiments of AFCNet with other deep-learning models are conducted. The baseline models, such as FCN [55], SegNet [3], Deeplabv3 [56], U-Net [4], TransUNet [5], Attention-UNet [6], Swin-UNet [7], and D-LinkNet [10], are compared on the both benchmarks. In addition, some recent works have been added to our comparison experiments, such as DCS-TransUperNet [13], ConDinet++ [11], Deep FR TransNet [54], SC-RoadDeepNet [51], AFU-Net [9], DDU-Net [2], and RADANet [57], respectively.

The visualized results on road extraction segmentation are shown in Figs. 7–9. It can be seen that AFCNet can successfully extract extremely weak and small-sized road objects from both datasets and performs better in identifying road objects from similar ones, such as rivers and gullies. The area marked by red rectangles in Fig. 7, and Fig. 8 is an instance of significant improvement in road segmentation by comparable methods. Furthermore, as shown in Fig. 9, the high-resolution visualized results in both datasets are depicted. It can be seen that our results generated more TPs and fewer FPs and FNs in homogeneous areas, which demonstrated our AFCNet performs well in discriminating small-sized road objects even when interfered with by the shadow of roadside vegetation.

The performance comparison is listed in Tables II and III. As shown in Table II on the Massachusetts dataset, AFCNet achieved the highest recognition accuracy of 79.34%, precision of 77.10%, recall of 75.65%, and mIoU of 83.15%, which

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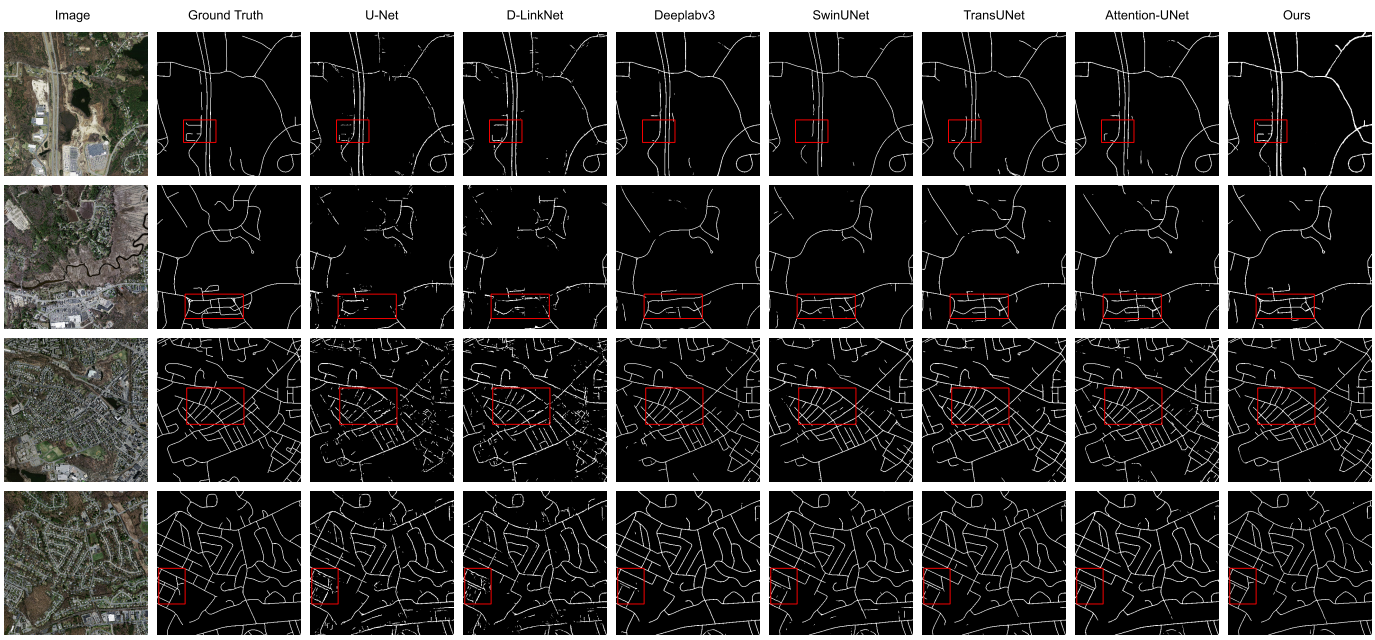


Fig. 7. Visualized results on the Massachusetts Road dataset. The first rows of images are urban traffic roads selected from the Massachusetts Road dataset.

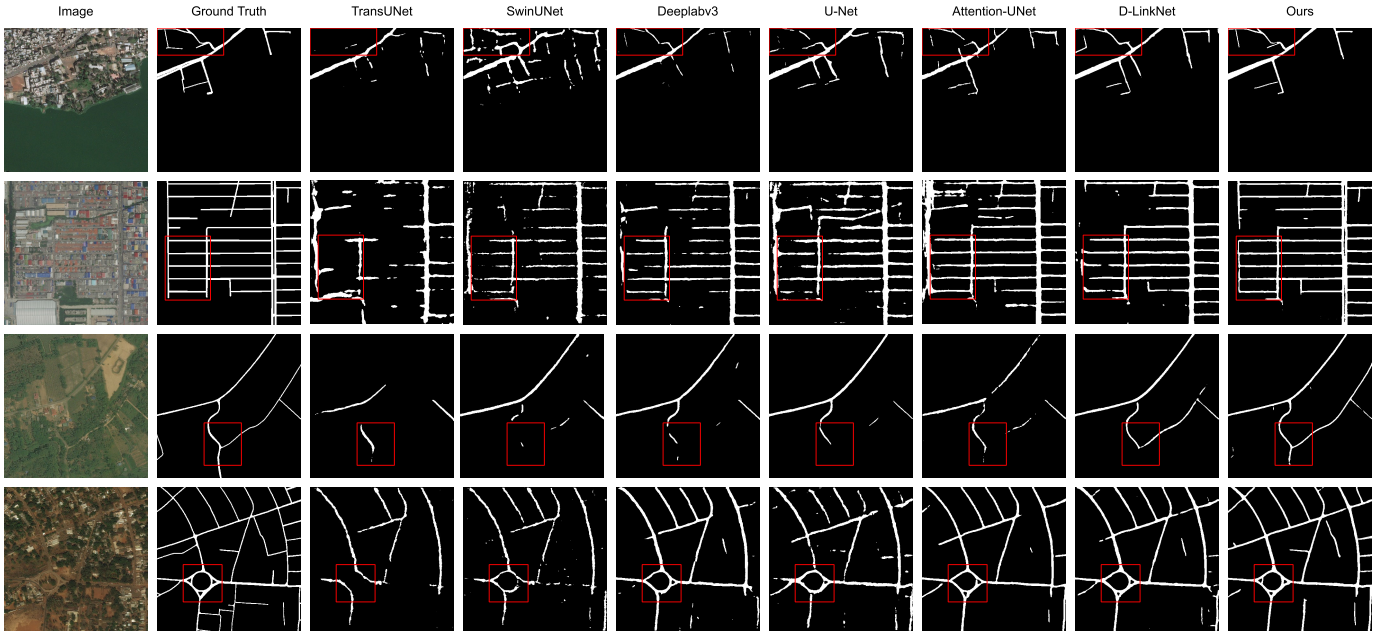


Fig. 8. Visualized results on the DeepGlobe dataset. The first rows of images are urban traffic roads selected from the DeepGlobe dataset.

obtained a 2.81%, 0.13%, 0.76%, and 0.46% gain compared with the SOTA method, that is, SC-RoadDeepNet [51]. It means AFCNet can accurately classify road pixels in RS images. Compared to the most recent Transformer model DCS-TransUperNet [13], AFCNet built on FNE demonstrates its superiority over the self-attention structure and outperforms it by 5.41% on mIoU. It improves mIoU score to 5.92%, 5.02%, 5.51%, and 6.76% compared with other comparative models, such as LinkNet [10], Attention-UNet [6], SWinUNet [7], and U-Net [4], respectively.

On the DeepGlobe dataset, it can be concluded from Table III that AFCNet shows enough competitiveness. Our AFCNet achieves the best accuracy of 98.89%, precision of 88.10%, recall of 81.81%, and mIoU of 78.26%, exceeding the recent methods (i.e., AFU-Net [9], DDU-Net [2], and DFC-UNet [58]) by 4.12%, 2.94%, and 1.54%, respectively, on mIoU. In contrast to classical convolutions and self-attentions, FNE has a larger receptive field based on the properties of Fourier Transform and has fine-grained feature extraction capability simultaneously. Compared to the most recent Transformer model Deep FR TransNet [54], AFCNet outperforms it by 5.82% on mIoU. Moreover, it improves mIoU score to 15.39%, 7.95%, 16.83%, and 17.24% compared with other comparative models, such

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TABLE II
RESULTS ON THE MASSACHUSETTS ROAD DATASET

Model	Accuracy (%) ↑	Precision (%) ↑	Recall (%) ↑	mIoU (%) ↑	Params (M) ↓	FLOPS (G) ↓	FPS ↑
FCN[57]	74.61	60.43	68.89	73.20	22.70	25.34	49
SegNet[3]	75.23	64.41	69.80	74.81	29.43	123.04	59
U-Net[4]	78.04	63.23	70.32	76.39	31.04	156.89	47
TransUNet[5]	78.10	68.35	71.23	77.89	67.62	99.51	50
Attention-UNet[6]	76.31	71.32	73.41	78.13	34.88	204.09	38
SwinUNet[7]	78.01	68.25	71.31	77.64	40.14	69.75	45
U-Net++[8]	78.66	65.90	71.27	76.77	-	-	-
Deeplabv3[58]	77.82	68.41	72.48	77.43	68.11	206.48	32
D-LinkNet[10]	77.83	67.16	71.73	77.23	52.65	63.69	60
DCS-TransUperNet[13]	77.94	69.52	73.48	77.74	-	-	-
ConDinet++[11]	78.14	72.05	74.71	78.88	-	-	-
AFU-Net[9]	78.93	76.83	75.17	80.32	110.04	191.34	14
DDU-Net[2]	98.04	82.54	73.99	80.98	-	-	-
RADANet[59]	79.11	77.91	73.91	81.65	-	-	-
SC-RoadDeepNet[53]	76.53	76.97	74.89	82.69	-	-	-
AFCNet (Ours)	**79.34**	**77.10**	**75.65**	**83.15**	**34.09**	**112.38**	**48**

TABLE III
RESULTS ON THE DEEPGLOBE ROAD DATASET

Model	Accuracy (%) ↑	Precision (%) ↑	Recall (%) ↑	mIoU (%) ↑	Params (M) ↓	FLOPS (G) ↓	FPS ↑
FCN[57]	92.05	63.71	61.54	42.45	22.70	25.34	49
SegNet[3]	91.54	62.03	60.21	41.85	29.43	123.04	59
U-Net[4]	97.12	82.28	71.20	61.02	31.04	156.89	47
TransUNet[5]	93.21	64.45	62.34	46.10	67.62	99.51	50
Attention-UNet[6]	96.79	79.89	79.47	70.31	34.88	204.09	38
SwinUNet[7]	97.01	82.35	71.54	61.43	40.14	69.75	45
Deeplabv3[58]	95.87	65.83	64.48	48.31	68.11	206.48	32
D-LinkNet[10]	98.32	83.33	72.29	62.87	52.65	63.69	60
Deep FR TransNet[56]	98.70	87.30	81.15	72.44	-	-	-
AFU-Net[9]	98.81	87.86	81.70	74.14	110.04	191.34	14
DDU-Net[2]	97.82	77.86	62.23	75.32	-	-	-
DFC-UNet[60]	98.08	83.26	79.14	76.72	-	-	-
AFCNet (Ours)	**98.89**	**88.10**	**81.81**	**78.26**	**34.09**	**112.38**	**48**

as LinkNet [10], Attention-UNet [6], SWinUNet [7], and U-Net [4], respectively.

It can be seen that the efficiency analysis shows that AFCNet achieved better performance with approximative computational costs. In Tables II and III, the FPS is measured by the time consuming of inference on each cropped subimage with a fixed size of 448×448.

Experimental results on both datasets demonstrate that AFCNet has better representation capability for road segmentation tasks. It can be concluded that our proposed AFCNet outperforms other comparable methods and can generate more accurate feature maps in RS images with cluttered backgrounds.

C. Ablation Study

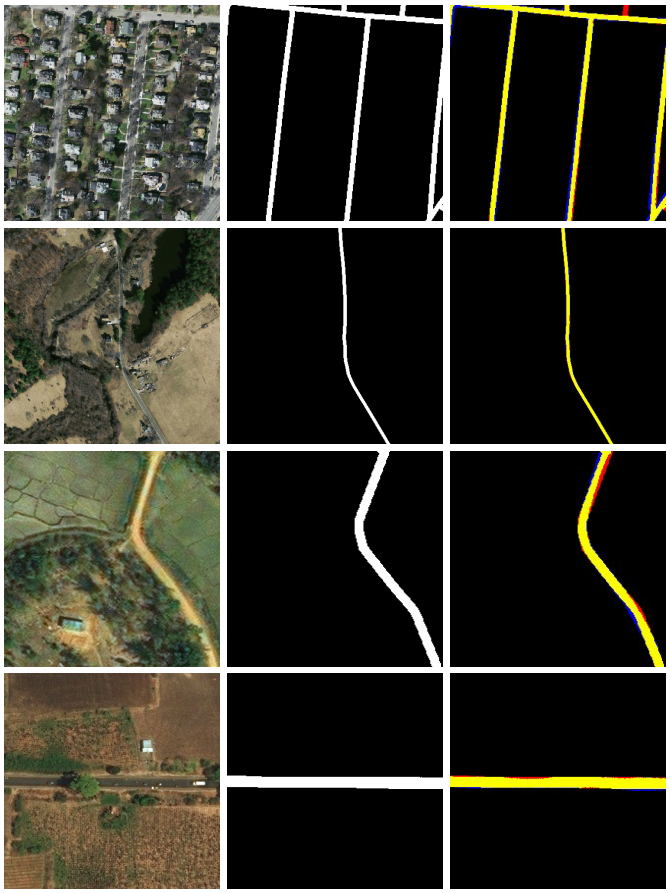
In this section, ablation studies on AFCNet model components, AFC modular structure, similar blocks comparison, and hyperparameters analysis are conducted to investigate the contribution of proposed layers and to explore the tradeoff on model performance and efficiency.

1) On Model Components: To evaluate the performance of the various modules of AFCNet, we took the classical U-Net [4] as a baseline and replaced all conventional convolution modules in U-Net with bottleneck in ResNet-50 (short for "Bottleneck") [43], or proposed FNE blocks (short for "FNE"), or hybrid encoders (marked with "Bottleneck + FNE") to perform an ablation study on the Massachusetts Road (short for "Mass. Road") and DeepGlobe datasets, respectively. The experimental results are listed in Table IV, which demonstrates that both components of FNE and bottleneck are critical to improving the performance of the AFCNet model.

We gradually replace the conventional convolution modules in U-Net with bottleneck alone, FNE alone, or part of bottleneck and part of FNE on both datasets. More specifically, after introducing bottleneck alone to the encoder or the decoder, they can boost the model by 2.25% and 13.60% on mIoU, respectively, and using 23.7M more parameters, and FNE modules alone can boost the baseline by 4.19% and 16.40% on mIoU, respectively, and consuming 17.24M fewer parameters. Nevertheless, when we use a hybrid encoder scheme in AFCNet as in Table I, the FNE modules are used for high-resolution representation, and bottleneck modules are used for low-resolution representation. The proposed hybrid encoder scheme can further boost the baseline to obtain 4.51% and 3.64% extra gains on mIoU on both datasets compared with the bottleneck-alone, while the hybrid encoder scheme only consumes comparable parameters with the baseline model. It can be concluded that the AFCNet model can achieve outstanding performance by benefiting from its FNE-based encoders.

2) On Modular Structure: We further investigate the inside of the FNE module and analyze the merits of the AFC module structure on Massachusetts Road and DeepGlobe

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(a) (b) (c)

Fig. 9. Visualized results of our proposed AFCNet model. (a) Original images from DeepGlobe and Massachusetts datasets. (b) Ground truth. (c) Our results. TPs, FPs, and FNs are represented by yellow, blue, and red, respectively.

TABLE IV
ABLATION STUDY ON MODEL COMPONENTS

Model	Params(M) ↓	mIoU(%) ↑	
		Mass. Road	DeepGlobe
U-Net (baseline) [4]	31.04	76.39	61.02
+ Bottleneck	54.74	78.64	74.62
+ FNE	13.80	80.58	77.42
+ Bottleneck+FNE	34.09	83.15	78.26

datasets. Several components, that is, ACNet core module (AC) [20], FAC, MA, and AMF in FNE modular in our AFCNet are deeply investigated to verify their contributions to the model's performance. The ablation study results are listed in Table V. Specifically, it can be seen that asymmetric convolution on the Fourier domain (marked with "FAC") overpasses the traditional ACNet by 2.29% and 10.76% on mIoU, respectively, by benefiting from the spectral convolution theorem [9], [15], which provides another path to capture global features for long and continuous road objects segmentation. Furthermore, when we gradually upgrade the FAC to AFC by adding MA and AMF mechanisms, the AFCNet can obtain 3.21% and 4.02% extra gain on mIoU on the Massachusetts Road dataset and obtain 3.40% and 3.64% extra gain on mIoU on the DeepGlobe dataset, respectively. Meanwhile, the parameters increase a bit at the cost. It can be concluded that fine-grained modulation learning on the

TABLE V
ABLATION STUDY ON FNE INSIDE

AC[20]	FAC	MA	AMF	Params(M) ↓	mIoU(%) ↑	
					Mass. Road	DeepGlobe
✓				11.46	76.84	63.86
	✓			32.68	79.13	74.62
	✓	✓		33.19	82.34	78.02
	✓	✓	✓	34.09	83.15	78.26

TABLE VI
ABLATION STUDY ON FNE INSIDE (CONTINUED)

Methods	Params(M) ↓	mIoU(%) ↑	
		Mass. Road	DeepGlobe
Skip connection	33.54M	82.35	77.22
Linear	34.09M	83.15	78.26
GAP standalone	34.09M	82.37	75.70
GMP standalone	34.09M	82.65	78.07
GAP+GMP	34.09M	83.15	78.26

Fourier domain by asymmetric convolutions plus MA and AMF mechanism is a fundamental layer for road segmentation from remote sensing images.

We further analyzed the structure of the FNE block to investigate the inside of the modular design. For instance, could the "linear" layer of FNE in Fig. 4(c) be replaced by a "skip connection"? Is the "GAP" standalone or "GMP" standalone enough for feature compression before the MA layer? We summarized and listed these exploration results in Table VI. From comparison analysis, it can be seen that the "linear" layer in our FNE modular can obtain 0.8% and 1.04% extra gains on both datasets, respectively, with a bit more parameters. The "GAP+GMP" mode in our FNE block can get 0.5%–0.78% and 0.19%–2.56% extra gains, respectively, compared with the GAP or GMP standalone mode.

3) Comparison With Similar Modules: Seen in Fig. 4, the proposed FNE block in this work originates from the Transformer encoder [30] and FNO [38], and in our AFCNet, we integrated the merits of these two blocks to form an AFC layer based on asymmetric Fourier convolution plus MA and AMF mechanisms. As we all know, self-attention-dominated Transformers [30] are powerful deep-learning models for most vision-related tasks. Among them, SwinUNet [7] and Deep FR TransNet [54] have achieved amazing results for road segmentation. Moreover, Fourier-based models, that is, FNO [38] and AFU-Net [9], have shown great potential for road segmentation. The results of the Massachusetts Road and DeepGlobe datasets from these similar models are summarized and listed in Table VII. It can be seen that AFCNet, meriting from both advantages of Transformer and FNO, outperformed those similar modules. Specifically, AFCNet outperforms the Transformer-based models [7], [30], [54] by 5.82%, 8.10%, and 16.83% on mIoU on the DeepGlobe and 5.51% and 7.72% on mIoU on the Massachusetts Road, respectively. Compared with previous Fourier-based models [9], [38], AFCNet achieves better performance and obtains 4.12%–4.57% extra mIoU gains on DeepGloble and obtains 2.83%–3.27% extra mIoU gains on Massachusetts Road, respectively. Compared with recent Fourier-related models [17], [19], although AFCNet needs more parameters,

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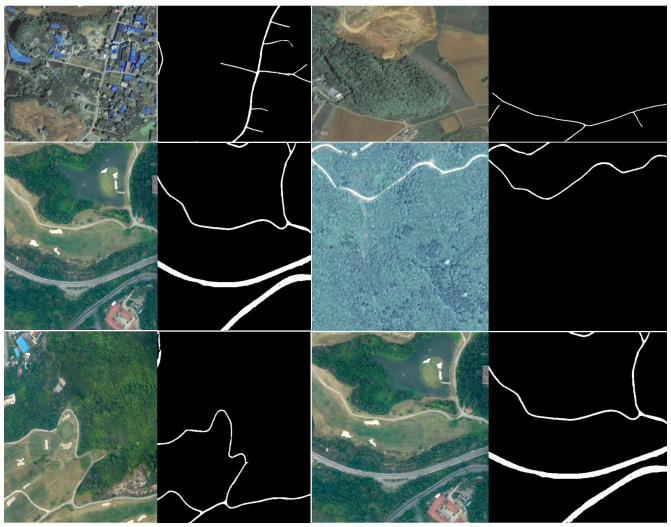


Fig. 10. Remote sensing images in the wild and the predicted results by our method.

TABLE VII
COMPARISON WITH SIMILAR MODULES

Encoder	Params(M) ↓	mIoU(%) ↑	
		Mass. Road	DeepGlobe
SwinUNet [7]	40.14	77.64	61.43
Transformer [31]	79.62	75.43	70.16
Deep FR TransNet [56]	-	-	72.44
FNO [39]	129.6	79.88	73.69
AFU-Net [9]	110.04	80.32	74.14
FIT [17]	8.32	67.81	45.72
FGCN [19]	0.81	64.95	55.44
AFCNet (Ours)	**34.09**	**83.15**	**78.26**

TABLE VIII
ABLATION STUDY ON THE CONVOLUTION KERNEL

Kernel size	mIoU(%) ↑	
	Mass. Road	DeepGlobe
(1, 3)	83.15	78.26
(1, 5)	82.97	77.93
(1, 7)	83.08	77.68

it overpasses those models on mIoU gain by 15.34%–18.2% on Massachusetts Road and 22.82%–32.54% on DeepGlobe, respectively. It is noticeable that AFCNet can achieve a better tradeoff on performance and parameters, which is more practical for many efficient scenarios.

4) On Convolution Kernel: To investigate the impact of asymmetric convolution kernel on model performance, we explored the optimal hyperparameter on kernel size on the Massachusetts Road and DeepGlobe datasets. For instance, we set the kernel with the size of (1, 3), (1, 5), and (1, 7), respectively, and the corresponding results are listed in Table VIII. These findings demonstrate that the kernel size of (1, 3) yields the best performance in our model.

5) On Loss Weight Analysis: We also explored how to choose the weight coefficient α in the loss function. When α is set to 0.05–0.25, and the mIoU obtained by our method on both datasets are listed in Table IX. Thus, $\alpha = 0.1$ is regarded as an optimized coefficient in our tasks.

TABLE IX
MIOU ON DIFFERENT LOSS WEIGHTS

Dataset	0.05	0.10	0.15	0.20	0.25
Mass. Road	83.03	83.15	83.12	82.99	82.95
DeepGloble	78.14	78.26	77.94	77.64	77.48

D. Robustness to Samples in the Wild

In real road segmentation applications, we usually encounter the problem of robustness to samples in the wild, where models are trained on public and labeled samples, but the samples in the wild to be predicted often fall out of the training set, and the model usually cannot work well for those wild remote sensing images with different backgrounds, contrasts, and road characteristics. Nevertheless, we tested the AFC model on some wild samples from the internet, and some selected visualized results are shown in Fig. 10. It can be seen that our model can perform well on those samples.

VI. CONCLUSION

In this work, we introduced an AFCNet, a new model for road segmentation from remote sensing images based on the FNO, asymmetric convolution, and Transformer. The proposed model follows the pipeline of classical U-Net and its main encoder and decoder modules consist of our proposed FNE and bottleneck in ResNet. The FNE originates from the Transformer encoder and a novel FNO block which is based on FAC plus MA and AMF mechanisms. The core encoder module of AFCNet combines the advantages of bottleneck block, FFN, FAC, MA, and AMF layers compactly and provides a significantly flexible and parameter-efficient global–local feature representation mechanism for road segmentation tasks. The experiments on two public datasets demonstrate the proposed AFCNet outperformed most previous models and similar models.

In the future, we will explore using the FNE block as a fundamental block for general feature representation learning for most vision-related tasks, improve its computational efficiency for real-time inference demands, and further put the FNE-based model to more applications.

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